High Resistivity Measurement of SiC Wafers Using Different Techniques

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To establish fast, nondestructive, and inexpensive methods for resistivity measurements of SiC wafers, different resistivity-measurement techniques were tested for characterization of semi-insulating SiC wafers, namely, the four-point probe method with removable graphite contacts, the van der Pauw method with annealed metal and diffused contacts, the current-voltage (I-V) technique, and the contactless resistivity-measurement method. Comparison of different techniques is presented. The resistivity values of the semi-insulating SiC wafer measured using different techniques agree fairly well. As a result, application of removable graphite contacts is proposed for fast and nondestructive resistivity measurement of SiC wafers using the four-point probe method. High-temperature van der Pauw and room-temperature Hall characterization for the tested semi-insulating SiC wafer was also obtained and reported in this work.

Key words: Resistivity, silicon carbide, four-point probe method, van der Pauw method, contactless-resistivity measurement

INTRODUCTION

Fast and nondestructive methods for the specification of the main SiC-wafer parameters are very important for providing fast feedback to the growth effort and SiC-wafer users. In Si technology, the resistivity is measured using four-point probe stations with pin contacts or different contactless methods. The very high resistance introduced by the contacts into the measurement circuit is the principle problem that makes the commercially available four-point probe systems with pin contacts extremely difficult to implement for SiC-wafer characterization. Reported resistivity measurements on SiC wafers using the van der Pauw method[1,2] is typically done on samples with annealed metal contacts, which is time consuming, and therefore, fast characterization response cannot be provided. Moreover, it was recently reported that annealing of SiC wafers decreases substrate bulk resistivity.[3,4] This aspect is crucial because the substrate is subjected to high temperatures during epitaxial growth of GaN or SiC films for the fabrication of microwave devices. Hence, for such applications, it is essential to determine the resistivity of the substrate after high-temperature annealing. Commercially available contactless resistivity gauges allow resistivity measurement in the range 0.001–100 Ω-cm on wafers with a minimum diameter of about 2 in. Although commercial systems for contactless resistivity measurements in a range 10^5–10^{11} Ω-cm do exist, these systems are expensive and, therefore, not widely used. Accounting for the preceding difficulties, the objective of this work was to establish a relatively inexpensive, fast, and nondestructive resistivity measurement method of SiC wafers including semi-insulating crystals based on the equipment widely available in the electrical characterization laboratory. One of the objectives was to avoid metal-contact deposition and annealing, which is a time consuming and inconvenient process.

The approach accented in this work for resistivity measurements is to use the four-point probe method with removable graphite contacts in square arrangement. The method was applied for high-resistivity SiC characterization. The results are cross-referenced with van der Pauw measurements with annealed metal and diffused contacts, current-voltage (I-V) measurements, and the contactless resistivity method using the impedance spectroscopy approach. High-temperature van der Pauw and Hall characterization for the tested semi-insulating wafer was also obtained and reported in this work.

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EXPERIMENTAL DETAILS

A commercial 4H semi-insulating Si-face wafer with a specified resistivity greater than 10^5 Ω-cm was used in the experiments. Note that the supplier did not specify the exact resistivity value.

To measure the resistivity in a given part of a wafer, without cutting it into pieces, the removable graphite contacts in a square arrangement (~6 mm × 6 mm) were made by applying graphite paint using a sharpened wood tip. The contacts were dried in air for about 5 min. The wafer with contacts was placed into the probe station (Model 6000, The Micromanipulator Co., Escondido, CA) and adjustable needle probes, with triaxial leads, were positioned on these contacts. The experimental schematic is shown in Fig. 1. Two Keithley electrometers (Model 6514) with input impedance greater than 2×10^{14} Ω were used for voltage measurements and a Keithley model 237 high voltage source measure unit was used to pass the current through the sample. Electrometer I in Fig. 1 acts as a unity gain amplifier, providing guarding and maximum insulation for the voltage terminal k, and Electrometer II senses the voltage between voltage terminals k and l. Because the guarding approach was used, it is believed that the input impedance of the voltage measuring part of the setup was as high as the electrometer itself, and the stray capacitance is minimal. The graphite contacts applied to the polished-SiC surface exhibit Schottky behavior. Therefore, in the case of graphite contacts, higher voltage than that for the annealed metal contacts may be needed to pass sufficient current through the sample. For the square contact arrangement away from the wafer edges, with a distance between contacts much larger than the wafer thickness, the resistivity can be evaluated using the formula

$$\rho = \frac{V_{k,l}}{I_{i,j}} \times \frac{2\pi w}{\ln(2)} \quad (1)$$

where ρ is the material resistivity, I is the applied current between contacts i and j, V is the measured voltage between contacts k and l, and w is the wafer thickness.

The samples with diffused- and annealed-metal contacts for the van der Pauw measurement were prepared as follows. The B-Al diffusion was accomplished via a graphite mask at 2,000°C for 10 min at 650 torr pressure of argon as a carrier gas.[5,6] Then, electron-beam vapor deposition of Al/Ni contacts was made at 100°C, following a rapid thermal annealing at 800°C for 10 sec in nitrogen. For the samples with annealed-metal contacts, the contacts were created by magnetron sputtering of Ni at 200°C through shadow mask followed by rapid thermal annealing at 800°C for 10 sec in nitrogen.

In the contactless method of the resistivity measurement, the impedance-frequency Z(ω) dependence for the capacitively coupled SiC wafer was used to determine the resistivity value. The schematic of the test fixture is shown in Fig. 2a, and the equivalent electrical circuit is shown in Fig. 2b. In this method, frequency dependence of the total impedance Z(ω)

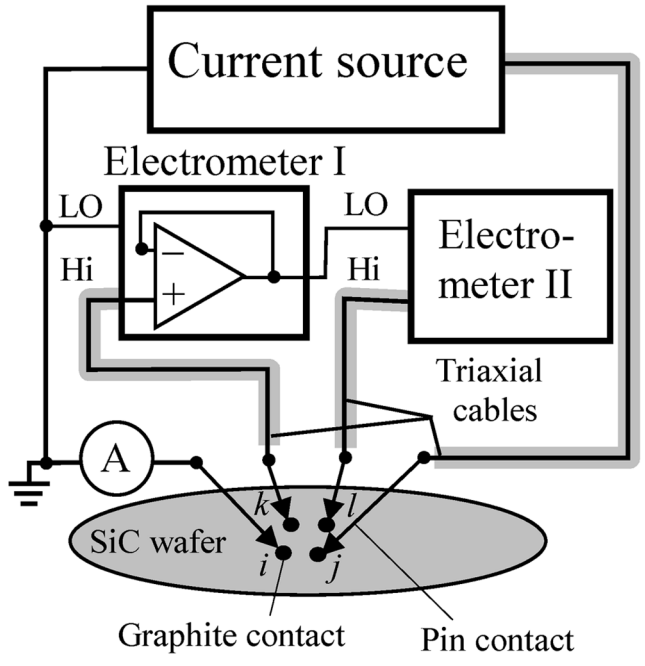


Fig. 1. The schematic of the four-point probe method with square contact arrangement using removable graphite contacts.

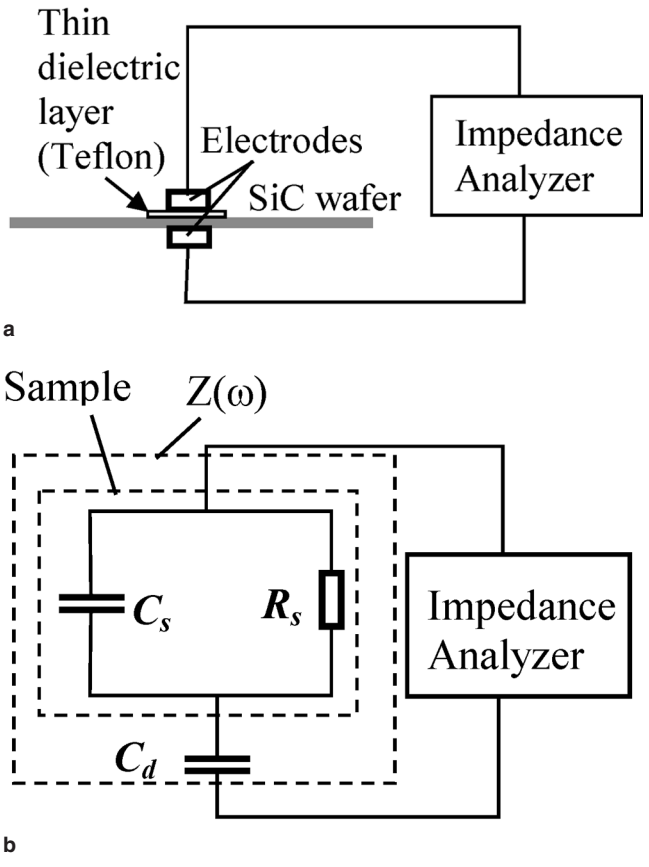


Fig. 2. (a) The experimental setup and (b) electrical diagram for the contactless-resistivity measurement.

was experimentally obtained using the Hewlett Packard model 4192A LF Impedance Analyzer (Hachioji-shi, Tokyo, Japan). For the circuit in Fig. 2b, the total impedance $Z(\omega)$ is given by

$$Z(\omega) = \frac{1}{j\omega C_d} + \frac{R_s}{1 + j\omega C_s R_s}, \qquad (2)$$

where $C_s = \varepsilon_o\varepsilon_s S/d_s$; $C_d = \varepsilon_o\varepsilon_d S/d_d$; $R_s = \rho d_s/S_s$; S is the electrode area; ε_s and ε_d are the dielectric constants of the semiconductor and the series dielectric layer, respectively; d_s, and d_d are the semiconductor and dielectric thickness, respectively; $\omega = 2\pi f$; f is the applied frequency; and ρ is the resistivity of the semiconductor wafer. Capacitances C_d and C_s were calculated from the wafer and dielectric layer thickness and geometry of the electrodes. Then, R_s and resistivity ρ were found by fitting the experimental and theoretical curves for $Z(\omega)$.

In the I-V method, both the lateral- and vertical-contact configurations were tested. The I-V characteristics using vertical-contact configuration with and without guard ring[7] were obtained to ensure that the measurements are not affected by the surface-leakage current.

RESULTS AND DISCUSSION

During resistivity measurements, we found strong dependence of the resistivity of the tested semi-insulating wafer to the room-light illumination and very large response time to the dark condition. The four-point probe and van der Pauw measurements indicate that under the room-light illumination, the resistivity of the sample was in the range 10^6–10^7 Ω-cm. After the sample was exposed to the dark, the resistivity increases at a rate of about 125 MΩ-cm/min as shown in Fig. 3. After an overnight period (~12 h), the resistivity of the sample typically was in the range 10^{10}–10^{11} Ω-cm. The observed increase of the resistivity in the dark evidences the presence of slow-trapping processes, which is believed to be caused by deep-level impurities. Hence, the SiC wafer under investigation is believed to exhibit semi-insulating nature because of compensation. Note that recent investigation showed that the semi-insulating SiC obtained using vanadium compensation exhibits a relaxation time to dark condition of about 10^3 sec in the order of magnitude.[8]

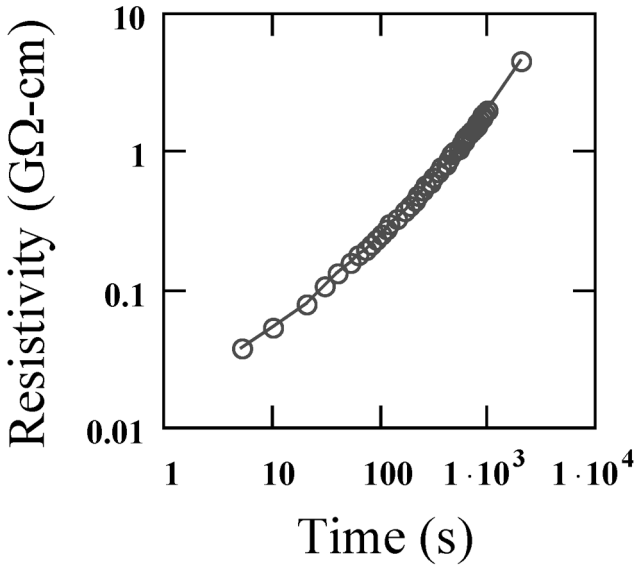


Fig. 3. The resistivity variation of the tested SiC sample with time in the dark.

Accounting for the strong resistivity dependence to light and dark conditions, the comparison between different techniques applied in this work was done at identical illumination or after the same time period the sample was left in the dark condition.

Figure 4 shows comparison of the current-voltage and van der Pauw characteristics obtained at room temperature under room-light illumination for the samples with annealed metal, diffused, and removable graphite contacts. As follows from Fig. 4a, annealed Ni and diffused contacts provide significantly lower contact resistance than that of graphite contacts. However, the large difference in contact resistance did not affect the van der Pauw characteristics (voltage between contacts k and l versus current between contacts i and j), as shown in Fig. 4b, which means that for all types of contacts the contact resistance stays less than the input impedance of our system. The resistivity values agree fairly closely for the samples with different types of contacts (1.2×10^6 Ω-cm and 9.4×10^5 Ω-cm for removable-graphite and diffused contacts, respectively). Thus, the preceding result proves that the use of temporary contacts is acceptable instead of metal contacts. It is important to note that application of removable contacts significantly decreases the time necessary for sample preparation for resistivity measurement, resulting in a fast and nondestructive method for wafer-resistivity measurement. However, because of the Schottky nature of the contacts, higher applied voltage is required for removable contacts, compared to the annealed-metal contacts, in order to pass sufficient current through the sample providing measured voltage above the noise limit.

To verify that the surface-leakage current did not affect our four-point probe and van der Pauw results the I-V characteristics of the sample from the same wafer with and without a guard ring were obtained (Fig. 5). The insert in Fig. 5 shows the schematic of the test structure. The resistivity calculated from the linear part of the forward I-V characteristic were 3×10^9 Ω-cm for the sample without guard ring kept for 40 min in the dark, 3.5×10^9 Ω-cm for the sample with guard ring kept for 40 min in the dark, and 1.4×10^{11} Ω-cm for the same sample with guard ring after overnight period in the dark. Thus, I-V characteristics and resistivity values indicate that the measured current is the bulk current, not the surface-leakage current. The resistivity values and time response to dark condition agree fairly well with those obtained using the van der Pauw and four-point probe methods.

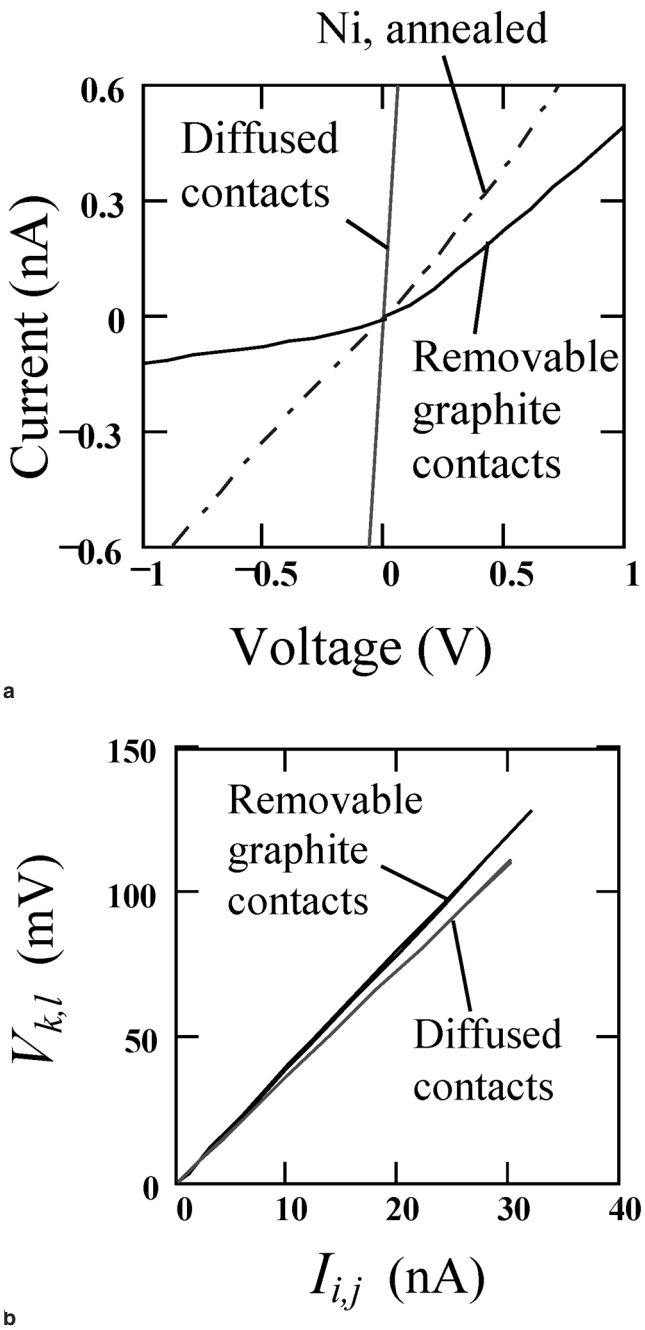


Fig. 4. The I-V and van der Pauw characteristics obtained for the samples with different types of contacts at room temperature under room-light illumination: (a) I-V characteristics for a given pair of contacts and (b) voltage between contacts k and l versus current between contacts i and j in the van der Pauw setup for the samples with removable and diffused contacts.

Figure 6 shows impedance versus frequency dependence obtained with the tested wafer. The best fit of the theoretical curve (solid line in Fig. 6) to the experimental data was obtained using the resistivity value 7×10^7 Ω-cm. Although the measurements were done under room-light illumination, the test area was between the electrodes, i.e., not directly exposed to room light. This fact explains the higher resistivity value obtained with this method compared

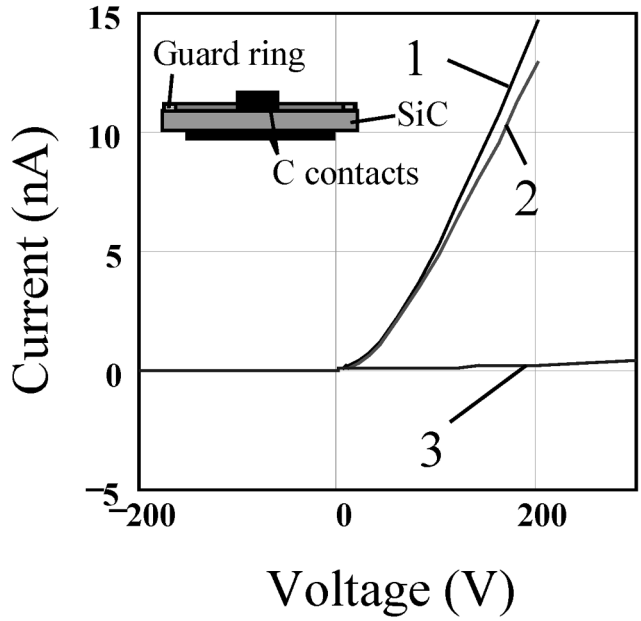


Fig. 5. The I-V characteristics for the vertical-contact configuration with and without guard ring. 1: No guard ring, 40 min in the dark (3×10^9 Ω-cm); 2: with guard ring, 40 min in the dark (3.5×10^9 Ω-cm); and 3: with guard ring after overnight in the dark (1.4×10^{11} Ω-cm).

to the van der Pauw results under room-light exposure. We were not able to measure the dark resistivity of the tested wafer using this method because the dark resistivity value exceeded the maximum resistivity value of about $\sim 10^8$ Ω-cm, which is possible to measure using our experimental setup for contactless-resistivity measurement.

Table I summarizes the results of resistivity measurements using different techniques for the samples from the same semi-insulating wafer. Values of resistivity obtained using different techniques agree fairly closely.

Finally, temperature dependence of the resistivity value was obtained under dark condition using the van der Pauw method with removable contacts (Fig. 7). The sample was kept in the dark condition for two days before the very first heating and the resistivity at room temperature was about 6×10^{11} Ω-cm as marked by number 1 in Fig. 7. At the very first heating of the sample, the resistivity drops from 6×10^{11} Ω-cm (room temperature) to about 3×10^{10} Ω-cm at 75°C, and then, as the temperature increases, the resistivity decreases according to line 2 in Fig. 7. Note that during the very first sample heating, at a given temperature, the resistivity slightly increases with time. During cooling of the sample from 620 K to room temperature, the resistivity follows line 3 in Fig. 7 with significantly higher resistivity, at a given temperature, than that during the very first heating. After cooling of the sample down to room temperature in the dark and subsequent heat up in the dark, the resistivity of the sample follows curve 3. Thus, the temperature dependence of the resistivity exhibits a memory effect to light illumination. Activation energies extracted from curve 2 (very first

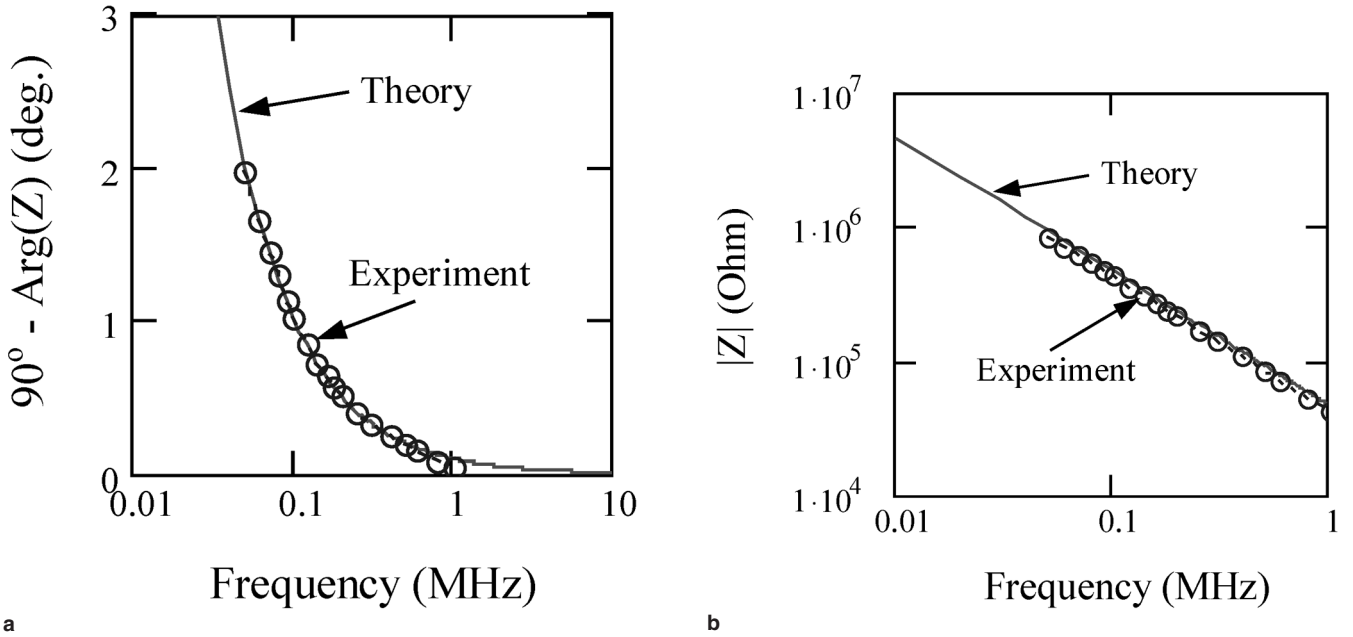


Fig. 6. The theoretical and experimental impedance versus frequency dependence of the tested semi-insulating wafer capacitively coupled to the measurement circuit (Fig. 2). Resistivity value of 7×10^7 Ω-cm gives best fit of the theoretical curve to the experimental data. —: theoretical fitting curve; and o: experimental data points.

heating) and curve 3 (cooling) were 0.202 eV and 1.21 eV, respectively. The activation energy of 1.21 eV indicates the presence of a deep-level impurity that is believed to act as a compensating impurity in the tested wafer. As follows from Fig. 7, the tested sample remained semi-insulating even under high-temperature conditions, which is an important result in considering the application of this material for high-temperature/high-power electronics.

Hall measurements on the tested semi-insulating wafer were affected by the presence of a large unbalanced voltage[7] and its continuous drift because of resistivity relaxation in the dark. After the overnight relaxation period, we were able to conduct room-temperature Hall measurements, resulting in a Hall coefficient of about 6.3×10^{11} cm^3/Coul, p-type conductivity, and 14 cm^2/Vs Hall mobility. We were not able to conduct high-temperature Hall measurements because of the noise produced by the heater. Further effort is needed for variable temperature Hall measurements on semi-insulating SiC wafers.

SUMMARY

Electrical characterization of high-resistivity SiC samples was obtained using different techniques. The different techniques employed gave consistent resistivity values, and the agreement in values is quite good. It was shown that it is possible to use removable contacts instead of metal-deposited contacts, which eliminates time-consuming contact deposition and contact-annealing steps, important for fast characterization of SiC wafers. Thus, the four-point probe method with removable contacts can be

Table I. Resistivity of the Samples Cut from the Same Semi-Insulating Wafer Measured by Different Techniques at Room Temperature

		Resistivity (Ω-cm)			
Method	**Contact Type**	**Room-Light Resistivity**	**Semidark***	**Dark Resistivity (Dark 40 min)**	**Dark Resistivity (Overnight in the Dark)**
van der Pauw	Ni, annealed	1.2×10^6	—	2.1×10^9	6×10^{10}
van der Pauw	Removable	9.4×10^5	—	3.1×10^9	1.7×10^{11}
van der Pauw	Diffused contacts	1.2×10^6	—	1.8×10^9	—
Four-point probe	Removable	4.4×10^6		3.4×10^9	—
I-V (vertical)	Removable	—	1.2×10^7	3.5×10^9	1.4×10^{11}
I-V (lateral)	Ni, annealed	1.6×10^6	—	2.7×10^9	8×10^{10}
Contactless Method	—	—	5×10^7	Out of range	Out of range

*Although testing is made under room light, the tested area is not exposed directly to room light (the tested area is located under the contact or in the test fixture).

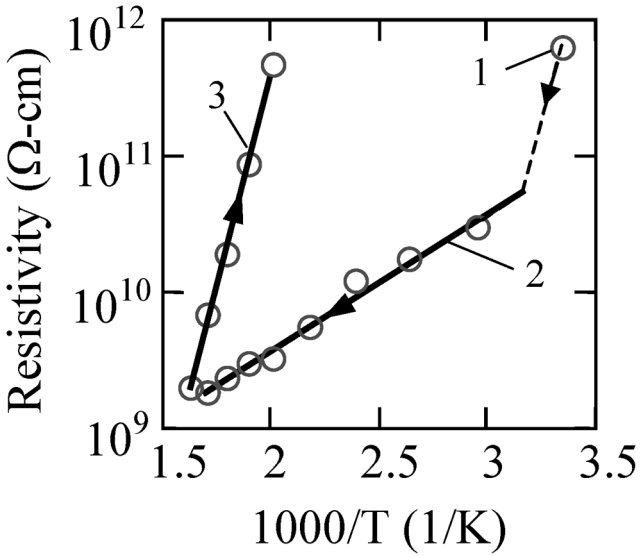


Fig. 7. The temperature dependence of the dark resistivity of the tested semi-insulating wafer. o: represents experimental data points; 1: marks the room-temperature resistivity of the sample that was in the dark for two days before the very first heating; 2: marks the resistivity dependence of the sample at the very first heat up in the dark from about 350–620 K; and 3: marks the resistivity curve during sample cool down in the dark from about 620 K.

used for fast and nondestructive resistivity measurements of high-resistivity SiC wafers. Investigations showed very strong sensitivity of the resistivity of the tested semi-insulating samples to visible light illumination and very large response time of the resistivity to the dark condition. It was shown that the tested semi-insulating samples remain semi-insulating even under high-temperature conditions, which reveals the high potential of these substrates to high-temperature/high-power applications. Presence of the energy levels with activation energies of 1.21 eV and 202 meV was revealed using high-temperature van der Pauw measurements.

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